

Topic 5. Effect of environmental regulation on Air Pollution and Infant Mortality in India (Paper: Hanna and Greenstone, American Economic Review, 2014)

Research questions for the extensions:

- **Empirical design:** What criticism would you make on the empirical strategy used by the authors? Consider the recent literature on two-way-fixed-effects (see paper by De Chaisemartin & D'Haultfoeille, Two-Way Fixed Effects and Differences-in-Differences with Heterogeneous Treatment Effects: A Survey) and what you have learnt about testing pre-trends.
- **Alternative design:** In light of the policy evaluation methods you have learnt in the “Microeconometric Evaluation of Public Policies” class, based on more recent papers, suggest and test an alternative method for estimating the effect of the catalytic converter policy and the supreme court action plan policy on air pollution and infant mortality. Write down the empirical specification, discuss the assumptions underlying a causal interpretation of the results, and compare your findings with the findings of the authors.

Available material

- **Datasets:**
 - o **combined.dta:** panel at the city and year level with information on city characteristics and air pollution policy intervention. Key variables
 - city: city name
 - year: year
 - e_spm_mean: Mean particulate matter measured in city that year
 - e_so2_mean : Mean so2 measured in city that year
 - e_no2_mean: Mean no2 measured in city that year
 - catconverter : dummy variable: 1 if catalytic converter policy is in place in that city that year
 - actionplan_sc: dummy variable: 1 if the city is under a mandate to develop an action plant that year
 - o **im_air.dta:** panel at the city and year level with information on infant mortality and air pollution policy intervention. Key variables:
 - city: city name
 - year: year
 - c_IM: infant deaths per thousand births
 - catconverter : dummy variable: 1 if catalytic converter policy is in place in that city that year
 - actionplan_sc: dummy variable: 1 if the city is under a mandate to develop an action plant that year
 - merge_air: 3 if observation for this year and city is also in the combined.dta dataset
- **Codes from the authors:** original do files used by the authors

- **Survey** of the recent literature on Two-Way-Fixed-Effects by De Chaisemartin and D'Haultfoeille (NBER Working Paper)